

Industry Solutions Report — Chikku Robotics

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1. Smart Manufacturing

Transform your factory with intelligent automation. Our cobots, AMRs, and AI vision systems work alongside your workforce to increase productivity, quality, and flexibility.

1.1 Assembly & Kitting

Collaborative robots handle repetitive assembly tasks with high precision, freeing human workers for higher-value activities. **CoBot C2 and C5** models are ideal for electronics, automotive, and consumer goods assembly lines.

1.2 Quality Inspection

AI-powered vision systems automatically detect defects, measure dimensions, and verify assembly correctness at production-line speeds. Our computer vision models achieve **99.7%+ defect detection rates** in electronics manufacturing.

1.3 Material Handling

CarryBot A1 AMRs autonomously transport raw materials, work-in-progress, and finished goods between production cells, warehouses, and shipping areas — eliminating manual forklift operations.

1.4 Machine Tending

CoBot arms load and unload CNC machines, injection molding presses, and test equipment 24/7. Quick changeover between product variants through teach-by-demonstration programming.

1.5 Packaging & Palletizing

End-of-line automation with **CoBot C5** for case packing and palletizing. Handles mixed-SKU pallets with AI-powered stacking optimization.

1.6 Case Study

A major electronics manufacturer deployed 12 **CoBot C2** units and 8 **CarryBot A1** AMRs across their PCB assembly line. Result: **40% throughput increase, 60% defect reduction, and ROI in 14 months.**

2. Healthcare & Medical

Our healthcare robotics solutions improve hospital efficiency, reduce staff burden, and enhance patient care — from logistics automation to AI-assisted diagnostics.

2.1 Hospital Logistics

MediBot M1 automates medication delivery, lab sample transport, linen distribution, and meal delivery throughout hospital campuses. Integrates with **HIS/EMR** for automated task dispatching.

2.2 Patient Assistance

ServeBot S2 deployed in patient rooms and wards for non-clinical assistance — delivering water, adjusting lights, calling nurses, and providing entertainment.

2.3 Disinfection

UV-C equipped mobile robots autonomously disinfect patient rooms, operating theaters, and common areas. Validated **99.99% pathogen elimination.**

2.4 Pharmacy Automation

CoBot arms handle medication picking, counting, and packaging in hospital pharmacies. Reduces medication errors and frees pharmacists for clinical duties.

3. Warehousing & Logistics

Automate your warehouse operations from receiving to shipping. Our AMRs and robotic systems handle goods movement, sorting, and order fulfillment with speed and accuracy.

3.1 Goods-to-Person Picking

CarryBot AMRs bring inventory shelves to pick stations, eliminating worker walking time. Increases pick rates from **80 to 300+ lines per hour**.

3.2 Cross-Docking & Sortation

Autonomous robots sort incoming goods by destination at high speed. **Reduces cross-dock processing time by 50%.**

3.3 Inventory Management

SkyBot I1 drones perform automated inventory counts in high-bay racking using barcode scanning and computer vision. **Counts 10× faster** than manual methods.

3.4 Last-Mile Delivery

SkyBot D1 drones deliver small parcels within a 15 km radius. **AutoHaul H1** autonomous vehicles handle larger deliveries on campus and between facilities.

4. Smart Cities & Public Infrastructure

Our robots contribute to safer, cleaner, and more efficient cities through autonomous inspection, security patrol, and public services.

4.1 Infrastructure Inspection

SkyBot I1 drones inspect bridges, tunnels, power lines, and buildings autonomously. AI detects cracks, corrosion, and structural issues early — preventing failures and **reducing inspection costs by 70%**.

4.2 Security & Surveillance

Autonomous mobile robots patrol public spaces, parking structures, and critical infrastructure. Anomaly detection AI alerts human operators to potential security threats.

4.3 Public Information

GuideBot G1 robots deployed in airports, train stations, and city centers provide wayfinding, event information, and multilingual assistance to citizens and tourists.

4.4 Environmental Monitoring

Sensor-equipped mobile robots monitor air quality, noise levels, and temperature across urban areas, providing real-time environmental data to city management systems.

5. Agriculture & Agritech

Precision agriculture powered by robotics and AI — from autonomous crop monitoring to selective harvesting.

5.1 Crop Monitoring

SkyBot I1 drones equipped with multispectral cameras survey crops, detecting disease, nutrient deficiency, and water stress before they become visible to the human eye.

5.2 Autonomous Weeding & Spraying

Mobile robots with computer vision identify and remove weeds mechanically or apply targeted spraying — **reducing herbicide use by up to 90%**.

5.3 Harvesting Assistance

CoBot arms with soft grippers and vision systems selectively harvest fruits and vegetables without damage. Adaptable to multiple crop types.

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